



Healthcare Data Science Research Internship

Data Visualization and Interpretation of Molecular Descriptors for Computational Drug Discovery

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WEEK 4 SUBMISSION

Data Visualization and Interpretation for Healthcare Insights

PROPOSED TOOLS & TECHNOLOGIES

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Data Visualization | Exploratory Data Analysis (EDA) | Chemical-Space
Analysis | Machine Learning

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Abstract

This project focuses on the visualization and interpretation of molecular descriptor data for computational drug discovery. A synthetic dataset representing chemical compounds was analysed using molecular descriptors including Molecular Weight (MW), LogP, Hydrogen Bond Acceptors (HBA), Hydrogen Bond Donors (HBD), Topological Polar Surface Area (TPSA), and Rotatable Bonds (RB), along with binary bioactivity classification.

The study followed a structured workflow involving data inspection, preprocessing, exploratory data analysis, and visualization. Different visualization techniques, including histograms, bar charts, scatter plots, box plots, correlation heatmaps, and pairwise analysis, were used to examine the distribution of molecular properties, relationships between descriptors, differences between active and inactive compounds, and patterns within the chemical-property space.

The visual findings were interpreted from a medicinal chemistry and computational drug discovery perspective, with particular attention to physicochemical properties such as molecular size, lipophilicity, polarity, hydrogen-bonding capacity, and molecular flexibility. The analysis also examined correlations and potential outliers that may be relevant to subsequent computational modelling.

The project demonstrates how molecular data visualization can support chemical data exploration, feature understanding, and preparation for machine learning and QSAR studies. However, because the dataset is synthetic and the bioactivity labels are not experimentally validated, the observed patterns should be considered computational observations rather than confirmed biological relationships.

Overall, the project provided practical experience in Python-based data visualization, molecular descriptor analysis, chemical-space interpretation, and the application of exploratory data analysis to computational drug discovery.

1. Introduction

Data visualization is an important component of modern healthcare research, pharmaceutical research, and computational drug discovery, as it enables complex datasets to be explored and interpreted more effectively. In drug discovery, large amounts of chemical and biological data are generated during the identification and optimization of potential drug candidates. Appropriate visualization techniques can help researchers identify patterns, relationships, distributions, trends, and unusual observations that may not be readily apparent from numerical data alone.

Molecular descriptors provide quantitative representations of important chemical and physicochemical characteristics of molecules. Descriptors such as Molecular Weight (MW), LogP, Hydrogen Bond Acceptors (HBA), Hydrogen Bond Donors (HBD), Topological Polar Surface Area (TPSA), and Rotatable Bonds (RB) are commonly used to characterize compounds. Understanding the distribution and relationships between these properties is relevant to medicinal chemistry because molecular size, lipophilicity, polarity, hydrogen-bonding ability, and molecular flexibility can influence important properties of drug-like molecules.

Visualization of molecular descriptor data can also support computational drug discovery and artificial intelligence-based approaches. Before developing machine learning or QSAR models, exploratory data analysis can help identify relationships between features, detect outliers, examine differences between compound groups, and assess the overall chemical-property space. This provides useful information for subsequent feature selection, model development, and interpretation.

The present project focuses on the visualization and interpretation of molecular descriptor data associated with bioactivity classification. A synthetic chemical dataset was analysed using Python-based data analysis and visualization techniques. The study examines molecular-property distributions, relationships between descriptors, differences between active and inactive compounds, correlations, and potential outliers. The resulting observations are interpreted from a medicinal

chemistry and computational drug discovery perspective, while recognizing the limitations associated with the use of a synthetic dataset.

The overall objective is to demonstrate how exploratory data analysis and molecular visualization can support the understanding of chemical datasets and provide a foundation for QSAR, machine learning, and AI-assisted drug discovery studies.

2. Healthcare Problem and Project Overview

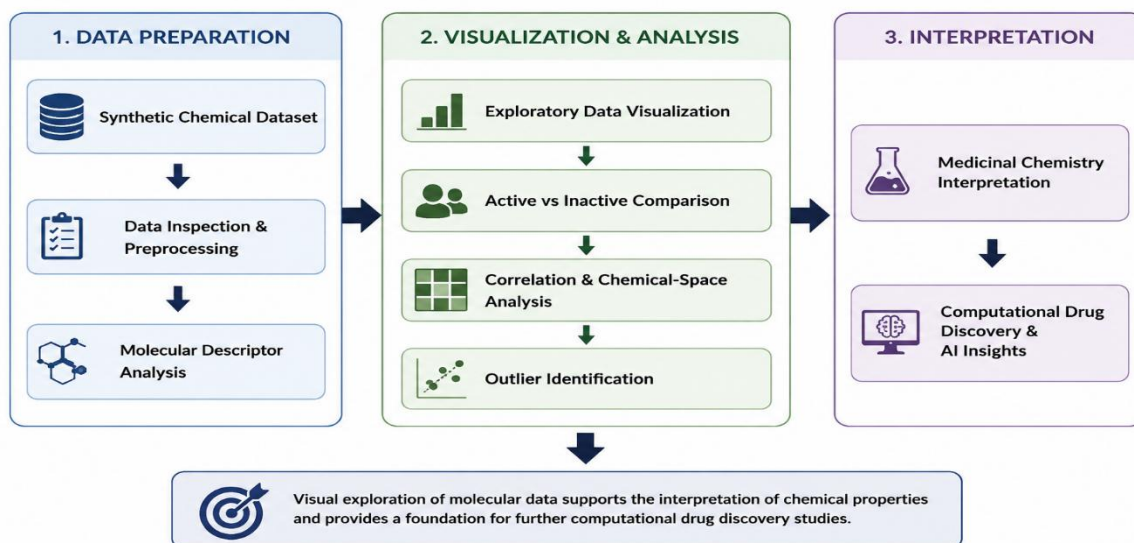
The identification and evaluation of promising chemical compounds are important stages in pharmaceutical and drug discovery research.

Chemical datasets often contain multiple molecular descriptors that describe the physicochemical characteristics of compounds. However, examining these properties only through numerical values can make it difficult to recognize patterns, relationships, differences between compound groups, and unusual observations. Data visualization provides an effective approach for exploring these characteristics and extracting meaningful insights from molecular datasets.

In this project, the healthcare-related problem is framed around the use of molecular-property visualization to explore patterns associated with the bioactivity of chemical compounds. The analysis aims to determine whether visualization of molecular descriptors can reveal differences between active and inactive compounds and identify relationships within the chemical-property space. Such exploratory analysis can provide useful preliminary information for subsequent computational approaches such as QSAR, machine learning, and AI-assisted drug discovery.

The project uses a synthetic bioactivity dataset containing molecular descriptors such as Molecular Weight, LogP, HBA, HBD, TPSA, and Rotatable Bonds, together with binary bioactivity classification. The dataset is first inspected and pre-processed before applying different visualization techniques, including histograms, bar charts, scatter plots, box plots, correlation heatmaps, and pairwise analysis.

The overall workflow is summarized below:



Overall Project Workflow

The project is intended to demonstrate how visual exploration of molecular data can support the interpretation of chemical properties and provide a foundation for further computational drug discovery studies. Since the dataset is synthetic, the findings are considered exploratory and should not be interpreted as experimentally validated biological conclusions.

3. Dataset Description and Data Assumptions

The dataset used in this project represents a collection of chemical compounds described using selected molecular and physicochemical properties. The dataset was designed to support exploratory visualization of molecular characteristics and their relationship with bioactivity classification. Since the dataset was synthetically generated, it was used primarily for demonstrating data visualization and interpretation techniques rather than for experimentally validated biological prediction.

3.1 Dataset Description

The dataset contains 500 chemical compounds, with each compound represented by a set of molecular descriptors and a binary bioactivity classification. The descriptors selected for analysis were:

Descriptor	Abbreviation	Description
Molecular Weight	MW	Represents the molecular mass of a compound
LogP	LogP	Indicates the lipophilic character of a compound
Hydrogen Bond Acceptors	HBA	Number of atoms capable of accepting hydrogen bonds
Hydrogen Bond Donors	HBD	Number of atoms capable of donating hydrogen bonds
Topological Polar Surface Area	TPSA	Represents the polar surface characteristics of a molecule
Rotatable Bonds	RB	Indicates molecular flexibility

The target variable, Bioactivity, was represented as a binary classification consisting of Active and Inactive compounds. These variables enabled comparisons between molecular-property distributions and bioactivity classes.

The dataset was generated and analysed using Python, with libraries such as Pandas, NumPy, Matplotlib, and Seaborn used for data handling and visualization.

3.2 Data Assumption

Because the dataset was synthetically generated, several assumptions were made for the purpose of this exploratory study:

- The molecular descriptors were assumed to represent meaningful physicochemical characteristics of chemical compounds.
- The dataset was assumed to contain sufficiently varied molecular properties for visualization and comparative analysis.
- The Active and Inactive labels were treated as binary bioactivity classes for analytical purposes.
- The descriptor values were treated as numerical variables suitable for statistical and graphical analysis.
- The synthetic dataset was assumed to be free from major data-entry errors after preprocessing.
- The observed relationships between descriptors and bioactivity were considered exploratory patterns rather than experimentally established biological relationships.

Therefore, the dataset provides a controlled environment for demonstrating molecular data visualization, chemical-space exploration, and medicinal chemistry interpretation, while its synthetic nature must be considered when interpreting the findings.

4. Data Preprocessing

Before performing visualization and exploratory analysis, the synthetic chemical dataset was inspected and prepared to ensure that the molecular descriptor data were suitable for analysis. Data preprocessing is an important step because inconsistencies, missing values, duplicate records, or inappropriate data types can affect the interpretation of visualizations and statistical relationships.

The dataset was first examined to understand its structure, variables, data types, and overall completeness. The molecular descriptors Molecular Weight (MW), LogP, HBA, HBD, TPSA, and Rotatable Bonds (RB) were

treated as numerical variables, while Bioactivity was treated as a categorical variable containing Active and Inactive classes.

The following preprocessing steps were performed:

- **Data inspection:** The dataset dimensions, column names, data types, and sample records were examined.
- **Missing-value assessment:** The dataset was checked for missing or null values in the molecular descriptor and bioactivity columns.
- **Duplicate assessment:** Duplicate records were examined to ensure that repeated observations did not unnecessarily influence the analysis.
- **Data-type verification:** Numerical descriptors were confirmed as numerical variables, while bioactivity was treated as a categorical variable.
- **Consistency checking:** Descriptor values were examined for inappropriate or unrealistic entries that could interfere with visualization.
- **Preparation for visualization:** The cleaned dataset was organized into a suitable format for generating distributions, comparisons, correlation plots, and chemical-space visualizations.

Since the dataset was synthetically generated, preprocessing primarily focused on data integrity and consistency rather than correcting experimentally measured chemical data. No experimentally validated values were introduced during preprocessing.

After preprocessing, the resulting dataset was used for molecular descriptor analysis and exploratory data visualization. The prepared data provided the basis for comparing active and inactive compounds, examining descriptor relationships, identifying potential outliers, and interpreting patterns from a medicinal chemistry and computational drug discovery perspective.

5. Molecular Descriptor Analysis

Molecular descriptors are numerical representations used to describe the structural and physicochemical characteristics of chemical compounds. In computational drug discovery, these descriptors provide a convenient way to represent molecules as quantitative features that can be explored through data analysis and subsequently used in approaches such as QSAR and machine learning.

In this project, six molecular descriptors were analysed: Molecular Weight (MW), LogP, Hydrogen Bond Acceptors (HBA), Hydrogen Bond Donors (HBD), Topological Polar Surface Area (TPSA), and Rotatable Bonds (RB). These descriptors were selected because they represent different aspects of molecular size, lipophilicity, polarity, hydrogen-bonding characteristics, and molecular flexibility.

Molecular Descriptor	Abbreviation	Relevance
Molecular Weight	MW	Represents molecular size and mass
LogP	LogP	Describes the lipophilic character of a compound
Hydrogen Bond Acceptors	HBA	Represents atoms capable of accepting hydrogen bonds
Hydrogen Bond Donors	HBD	Represents atoms capable of donating hydrogen bonds
Topological Polar Surface Area	TPSA	Provides an estimate of molecular polarity
Rotatable Bonds	RB	Provides an indication of molecular flexibility

Molecular Weight (MW) provides information about the overall size of a molecule and can influence properties such as molecular transport and physicochemical behaviour. LogP represents the lipophilic character of a compound and is an important property when considering interactions between compounds and different chemical environments.

HBA and HBD describe the hydrogen-bonding capabilities of molecules and can contribute to molecular recognition and interactions with biological targets. TPSA provides information about the polar surface characteristics of a molecule, while Rotatable Bonds provide an indication of molecular flexibility.

The distribution and relationships of these descriptors were explored using different visualization techniques in the subsequent sections. Examining these properties together helps provide a broader understanding of the chemical-property space represented by the dataset and supports interpretation from a medicinal chemistry and computational drug discovery perspective.

It should be noted that individual molecular descriptors do not independently determine biological activity. Their interpretation should therefore be considered in combination with other molecular and biological information.

6. Exploratory Data Visualization

Exploratory data visualization was performed to examine the distributions, relationships, and patterns present within the molecular descriptor dataset. Visualization provides a useful approach for converting numerical molecular-property data into graphical representations that can be interpreted more easily from a medicinal chemistry and computational drug discovery perspective.

Different visualization techniques were selected according to the type of information being investigated. Histograms were used to examine the distributions of individual molecular descriptors, while a bar chart was used to visualize the distribution of bioactivity classes. Scatter plots were

used to investigate relationships between pairs of molecular descriptors and identify possible trends or unusual observations.

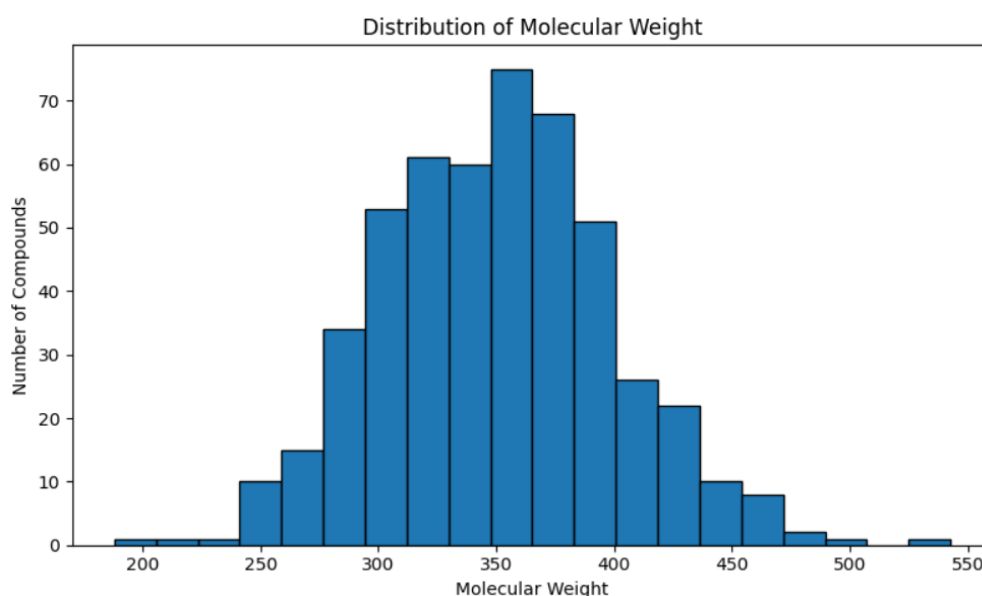
The analysis focused on the following visualizations:

- Molecular Weight distribution
- LogP distribution
- Bioactivity class distribution
- Molecular Weight vs LogP
- TPSA vs LogP

These visualizations provide an initial understanding of the chemical-property space represented by the dataset and help identify patterns that can be examined in greater detail in later sections. The visual results are interpreted based on the observed data patterns rather than being treated as evidence of direct causal relationships with biological activity.

6.1 Molecular Weight Distribution

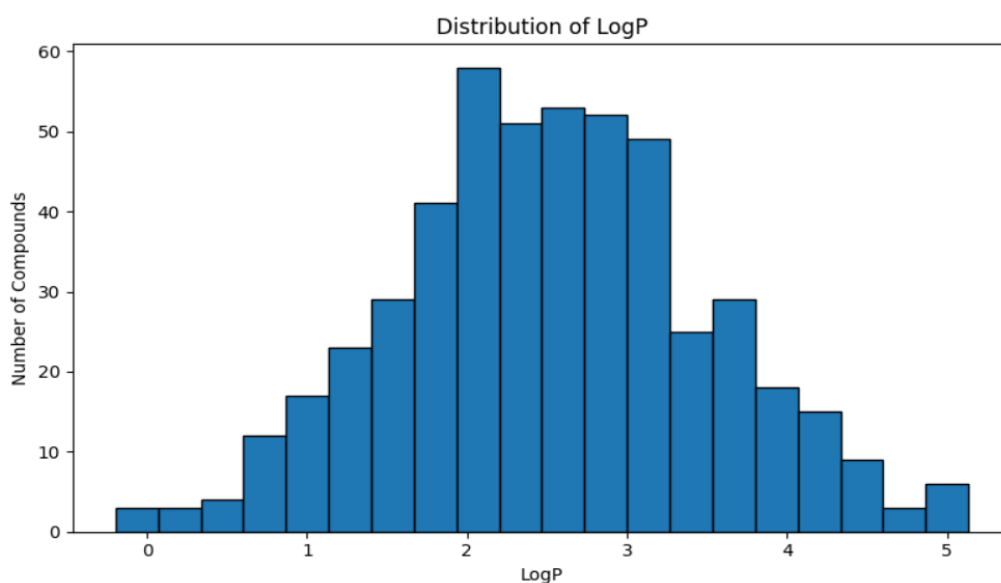
A histogram was used to visualize the distribution of Molecular Weight (MW) across the 500 chemical compounds. This visualization helps examine the range and concentration of molecular sizes within the dataset and identify potential unusually high or low molecular-weight observations.



Distribution of Molecular Weight among the chemical compounds

6.2 LogP Distribution

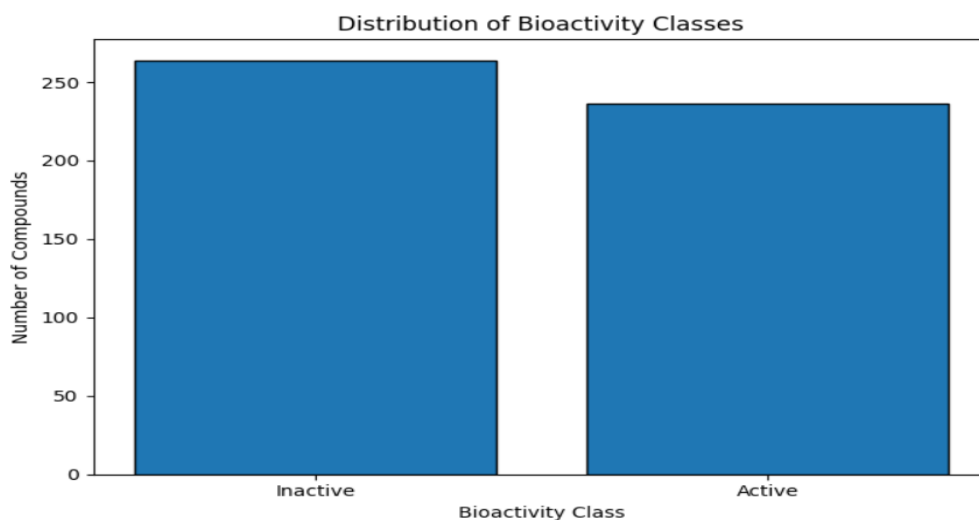
The distribution of LogP values was examined using a histogram. LogP provides an indication of the lipophilic character of the compounds. Visualizing its distribution helps examine the range of lipophilicity represented in the dataset and identify potential unusually high or low values.



Distribution of LogP values among the chemical compounds

6.3 Bioactivity Class Distribution

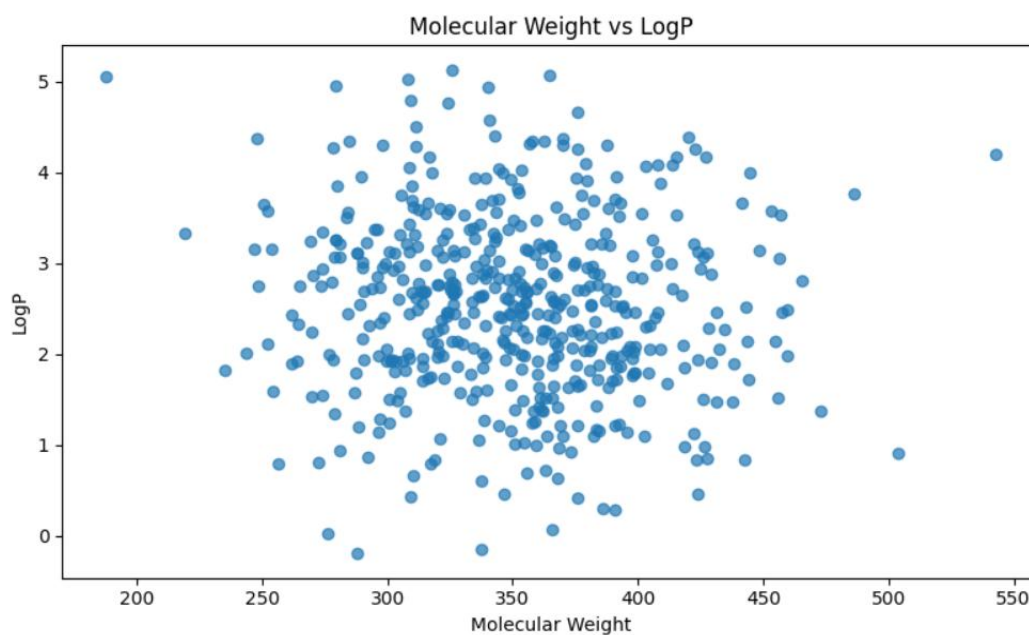
A bar chart was used to compare the number of Active and Inactive compounds in the dataset. Examining the class distribution is important because an imbalance between classes may influence subsequent comparative analysis and machine learning applications.



Distribution of Active and Inactive chemical compounds

6.4 Molecular Weight vs LogP

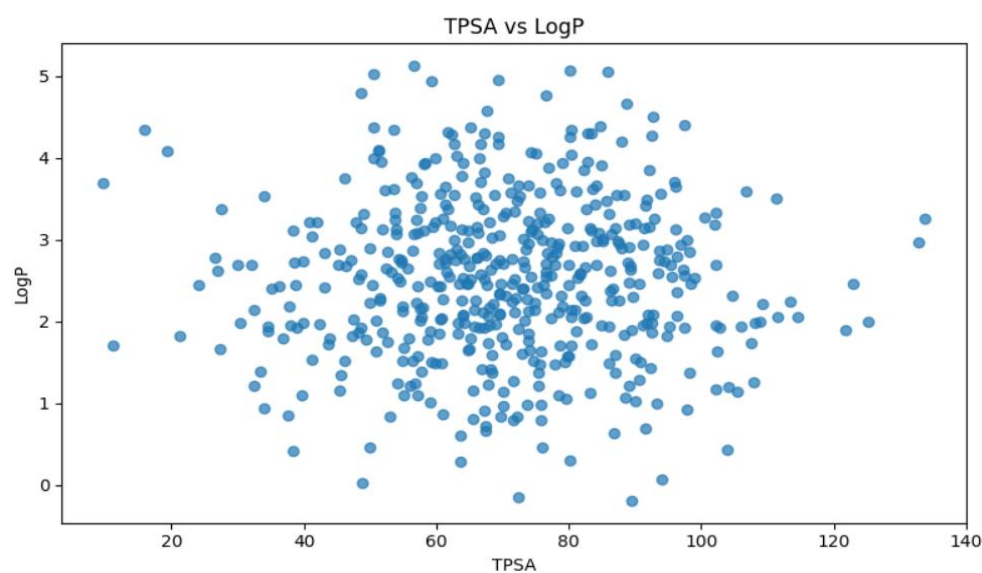
A scatter plot was used to examine the relationship between Molecular Weight and LogP. This visualization allows the distribution of compounds across molecular size and lipophilicity to be observed simultaneously. It can also help identify potential trends, patterns, or unusual observations within the chemical-property space.



Relationship between Molecular Weight and LogP

6.5 TPSA vs LogP

A scatter plot was used to investigate the relationship between Topological Polar Surface Area (TPSA) and LogP. These properties represent different aspects of molecular polarity and lipophilicity. Visualizing them together provides an initial understanding of how compounds are distributed across these two physicochemical dimensions.



Relationship between TPSA and LogP

Overall, these visualizations provide the first level of exploratory analysis of the molecular dataset. More detailed comparisons between Active and Inactive compounds, as well as correlation and chemical-space analysis, are presented in the following sections.

7. Comparative Analysis of Active and Inactive Compounds

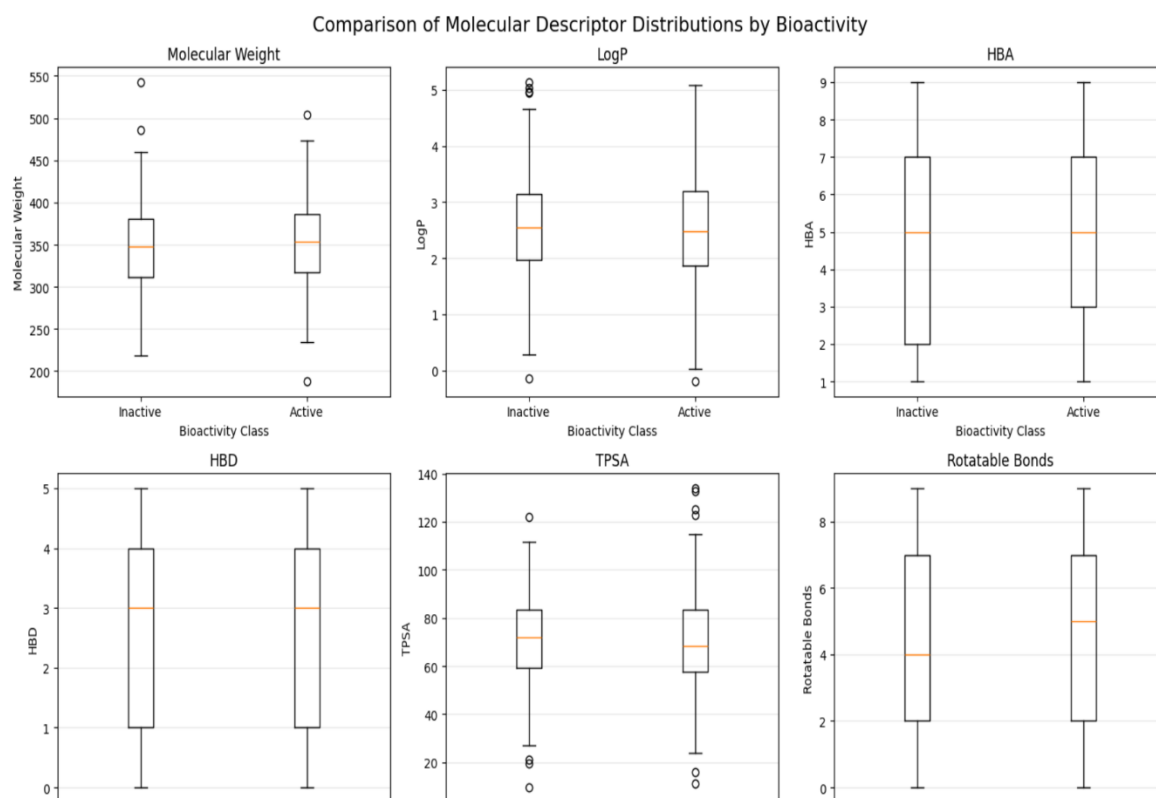
The molecular properties of Active and Inactive compounds were compared to identify differences in their physicochemical characteristics. Comparing the two bioactivity classes provides a more detailed view of how molecular descriptors are distributed across the dataset and helps identify properties that may show noticeable differences between the two groups.

The analysis focused on Molecular Weight (MW), LogP, HBA, HBD, TPSA, and Rotatable Bonds (RB). Box plots were used to compare the distributions of these descriptors between Active and Inactive compounds. This approach allows differences in the median, spread, and potential outliers of each molecular property to be visually examined.

The comparison is exploratory because the dataset is **synthetically generated** and the bioactivity labels are not experimentally validated. Therefore, any differences observed between the two classes should not be interpreted as evidence that a particular molecular descriptor directly causes biological activity.

7.1 Distribution of Molecular Properties by Bioactivity

The distributions of the selected molecular descriptors were compared between Active and Inactive compounds using graphical analysis. The comparison provides an overview of whether the two bioactivity classes occupy similar or different regions of the molecular-property space.



Comparison of molecular descriptor distributions between Active and Inactive compounds

The visualization compares the following properties:

- Molecular Weight (MW)
- LogP
- Hydrogen Bond Acceptors (HBA)
- Hydrogen Bond Donors (HBD)
- Topological Polar Surface Area (TPSA)
- Rotatable Bonds (RB)

Differences in the distributions may indicate that certain molecular properties vary between the two classes. However, substantial overlap between the distributions would indicate that the individual descriptors alone may not clearly distinguish Active from Inactive compounds.

7.2 Interpretation

The comparative analysis provides an initial understanding of the physicochemical characteristics associated with the two bioactivity classes. If a descriptor shows noticeably different distributions between Active and Inactive compounds, it may contain useful information for subsequent computational analysis.

However, molecular descriptors should not be interpreted independently. Bioactivity is influenced by multiple structural, physicochemical, and biological factors, and the selected six descriptors provide only a limited representation of chemical structure.

The observed patterns can therefore be considered useful for exploratory medicinal chemistry analysis and feature understanding, but they should not be treated as experimentally validated structure–activity relationships. Further analysis using molecular fingerprints, additional descriptors, experimentally measured bioactivity data, and appropriate machine learning or QSAR methods would be required to establish stronger predictive relationships.

8. Correlation and Chemical-Space Analysis

Correlation and chemical-space analysis were performed to further investigate the relationships among the molecular descriptors in the dataset. While the previous sections examined individual descriptor distributions and comparisons between Active and Inactive compounds, correlation analysis provides a broader view of how the selected physicochemical properties vary in relation to one another.

The analysis focused on the six molecular descriptors: Molecular Weight (MW), LogP, HBA, HBD, TPSA, and Rotatable Bonds (RB). A correlation heatmap and pairwise descriptor analysis were used to identify the strength and direction of linear relationships between these variables.

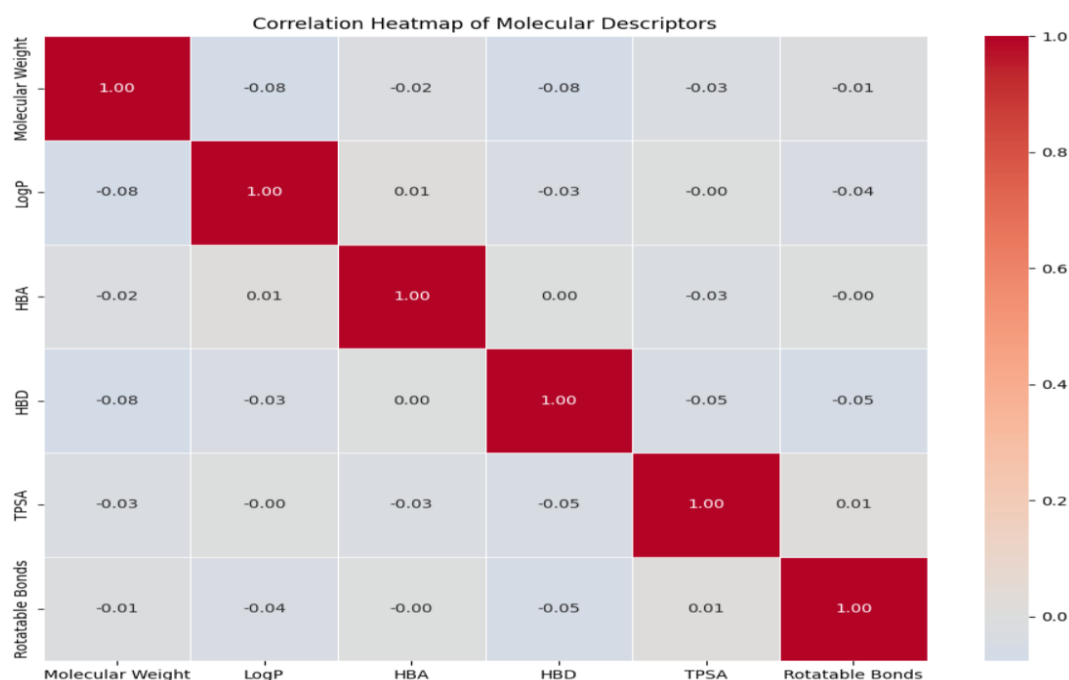
Understanding these relationships is relevant to medicinal chemistry and computational drug discovery, as correlated molecular descriptors may provide overlapping information and can influence the interpretation of subsequent statistical, QSAR, and machine learning analyses.

8.1 Correlation Heatmap

A correlation heatmap was generated to visualize the pairwise correlations among the six molecular descriptors. The heatmap provides a compact representation of the strength and direction of linear relationships between the variables.

Correlation coefficients range from -1 to $+1$. A positive value indicates that two descriptors tend to increase together, whereas a negative value indicates an inverse relationship. Values close to zero indicate a relatively weak linear relationship.

The heatmap was used to identify descriptors that show relatively strong or weak relationships within the synthetic chemical dataset. Strong correlations may indicate that certain descriptors contain related information, whereas weak correlations suggest that the descriptors provide more independent information.

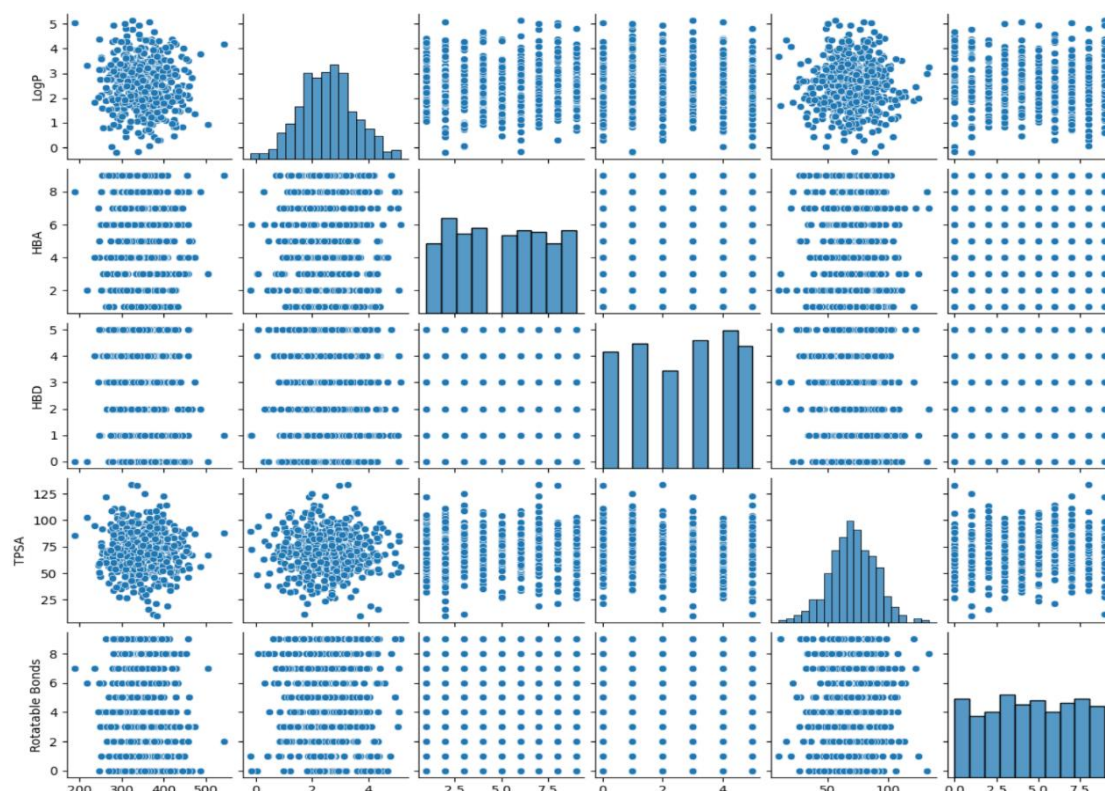


Correlation heatmap of molecular descriptors

8.2 Pairwise Descriptor Analysis

Pairwise analysis was used to examine relationships between multiple molecular descriptors simultaneously. Scatter plots allow the distribution and relationship between pairs of descriptors to be visually assessed, while diagonal distributions provide information about the individual variables.

The pairwise analysis complements the correlation heatmap by providing a visual representation of the relationships identified through correlation coefficients. It can also help reveal potential trends, unusual observations, and differences in the overall chemical-property space.



Pairwise relationships among selected molecular descriptors

8.3 Chemical-Space Interpretation

The combined correlation and pairwise analyses provide an overview of the chemical-property space represented by the dataset. Molecular compounds can occupy different regions of this space depending on their molecular size, lipophilicity, polarity, hydrogen-bonding characteristics, and flexibility.

The analysis helps identify whether some molecular properties vary together and whether the dataset contains broad or concentrated regions of chemical-property space. Such information can be useful during exploratory analysis and can provide context for subsequent QSAR, machine learning, and AI-assisted drug discovery studies.

However, the correlations should be interpreted cautiously, as correlation does not imply causation. Since the dataset is synthetic, the findings are exploratory and intended to demonstrate chemical-space analysis and molecular descriptor interpretation.

9. Outlier Analysis

Outlier analysis was performed to identify chemical compounds with molecular descriptor values that differ substantially from the majority of observations. Identifying such observations is important in molecular data analysis because extreme values may influence statistical relationships and visualizations.

Potential outliers were examined using the box plots presented in Fig. Correlation heatmap of molecular descriptors. and the distributions generated during exploratory data analysis. The analysis considered Molecular Weight, LogP, HBA, HBD, TPSA, and Rotatable Bonds.

The presence of an outlier does not necessarily indicate an error. In chemical datasets, unusual molecular properties may represent compounds with genuinely different structural or physicochemical characteristics. Therefore, potential outliers were considered part of the chemical-property space rather than automatically removed.

Since the dataset is synthetically generated, extreme observations were interpreted cautiously. No observations were removed solely because they appeared as potential outliers. Instead, they were retained for visualization and subsequent analysis to preserve the characteristics of the generated dataset.

Overall, outlier analysis helped identify the range and variability of molecular properties and provided additional context for interpreting the chemical-property space from a medicinal chemistry and computational drug discovery perspective.

10. Medicinal Chemistry Interpretation

The molecular descriptor analysis provides a useful perspective for understanding the physicochemical characteristics of the chemical compounds in the dataset. From a medicinal chemistry perspective, properties such as molecular size, lipophilicity, polarity, hydrogen-bonding

ability, and molecular flexibility can influence how compounds behave and interact with biological systems.

10.1 Physicochemical Property Patterns

The analysis of Molecular Weight, LogP, HBA, HBD, TPSA, and Rotatable Bonds provides an overview of the physicochemical characteristics represented in the dataset. The distributions and comparisons between Active and Inactive compounds show substantial overlap across several descriptors, indicating that individual properties do not clearly distinguish the two bioactivity classes.

These properties should therefore be considered collectively rather than independently. Molecular size and lipophilicity can influence molecular interactions and physicochemical behaviour, while hydrogen-bonding properties and TPSA provide information about polarity and potential molecular recognition. Rotatable Bonds provide an indication of molecular flexibility.

10.2 Implications for Drug Discovery

Understanding molecular-property patterns is important during early-stage drug discovery and medicinal chemistry, where compounds are evaluated based on multiple physicochemical characteristics. Visualization of these properties can help researchers explore chemical space, identify unusual compounds, and select molecular features for subsequent computational analysis.

In this project, the observed patterns provide a preliminary foundation for approaches such as QSAR, molecular fingerprint analysis, machine learning, and AI-assisted drug discovery. However, the synthetic nature of the dataset and the absence of experimentally validated bioactivity data mean that these observations should be treated as exploratory rather than confirmed structure–activity relationships.

11. Relevance to Computational Drug Discovery and AI

The analysis of molecular descriptors and their visualization is directly relevant to computational drug discovery and AI-assisted pharmaceutical research. Molecular descriptors convert chemical structures into numerical features that can be analysed computationally, making them useful for exploratory analysis, QSAR modelling, and machine learning applications.

11.1 Molecular Descriptors and Machine Learning

Molecular descriptors such as Molecular Weight, LogP, HBA, HBD, TPSA, and Rotatable Bonds can be used as input features for machine learning models. Visualization and correlation analysis performed in this project help researchers understand the distribution of these features, identify relationships between them, and recognize potential outliers before model development.

The findings from exploratory analysis can therefore support important data preprocessing and feature-understanding steps in computational workflows. However, the six descriptors provide only a limited representation of molecular structure, and additional molecular descriptors or fingerprints may be required for more comprehensive modelling.

11.2 QSAR and AI-Based Drug Discovery

The analysis provides a foundation for Quantitative Structure–Activity Relationship (QSAR) studies, where molecular features are related to experimentally measured biological activity. Machine learning and AI methods can further utilize molecular descriptors and structural fingerprints to identify patterns within chemical datasets and support the prioritization of compounds for further investigation.

In a practical drug discovery workflow, these approaches can complement experimental research by helping researchers explore chemical space,

prioritize compounds, and generate testable hypotheses. The present project demonstrates the exploratory data-analysis stage of such a workflow. Since the dataset is synthetic and its bioactivity labels are not experimentally validated, the results should not be interpreted as an actual AI-based prediction of drug activity.

Overall, this analysis demonstrates how molecular descriptors, visualization, QSAR, and AI-based methods can contribute to modern computational drug discovery workflows.

12. Key Findings and Critical Evaluation

The project demonstrated how data visualization and exploratory analysis of molecular descriptors can be used to understand the chemical-property space of a synthetic bioactivity dataset. The major findings and limitations identified during the analysis are summarized below.

Key Findings

- The dataset contained 500 synthetic chemical compounds described using Molecular Weight, LogP, HBA, HBD, TPSA, and Rotatable Bonds.
- The distributions of molecular descriptors provided an overview of the physicochemical characteristics and chemical-property space represented in the dataset.
- Comparisons between Active and Inactive compounds showed substantial overlap across several molecular descriptors, indicating that individual descriptors did not clearly separate the two bioactivity classes.
- Correlation and pairwise analyses helped identify relationships among the molecular descriptors and provided a broader view of the chemical-property space.
- Outlier analysis highlighted compounds with unusual descriptor values without automatically treating them as errors.

- From a medicinal chemistry perspective, the analysis demonstrated the importance of considering multiple molecular properties collectively rather than relying on a single descriptor.
- The workflow provides a useful foundation for further computational approaches such as QSAR, molecular fingerprint analysis, machine learning, and AI-assisted drug discovery.

Critical Evaluation

The analysis successfully demonstrated the application of Python-based visualization techniques to molecular descriptor data. However, the dataset was synthetically generated, and the bioactivity labels were not experimentally validated. Therefore, the observed patterns cannot be considered confirmed biological relationships or experimentally validated structure–activity relationships.

Additionally, the six molecular descriptors provide only a limited representation of chemical structure. More comprehensive computational studies would require additional descriptors, molecular fingerprints, experimentally measured bioactivity data, and appropriate validation strategies.

Overall, the project demonstrates how exploratory visualization can support molecular data interpretation and provide a starting point for computational drug discovery workflows, while also highlighting the importance of careful interpretation and data quality.

13. Recommendations

Based on the visualization and exploratory analysis performed in this project, the following recommendations can be considered for future computational drug discovery studies:

- Use experimentally validated datasets: Future analyses should use experimentally measured molecular structures and bioactivity values to obtain more reliable conclusions.

- Include additional molecular features: Molecular fingerprints, structural descriptors, and other physicochemical properties can provide a more comprehensive representation of chemical structure.
- Apply QSAR and machine learning: The analysed descriptors can be used as features for developing and evaluating QSAR and machine learning models.
- Perform appropriate feature selection: Correlation analysis and other feature-selection methods can help identify informative descriptors and reduce redundant information.
- Validate models carefully: Future predictive studies should use cross-validation, independent test sets, and multiple performance metrics to assess model generalization.
- Expand chemical-space analysis: Larger and chemically diverse datasets can provide a broader representation of molecular property space.
- Integrate experimental and computational approaches: Computational predictions should be combined with laboratory validation before drawing conclusions about biological activity.
- Use AI as a complementary tool: AI and machine learning can assist in compound prioritization and hypothesis generation, but their predictions should be interpreted alongside chemical and biological evidence.

Overall, future work should progress from exploratory visualization → molecular descriptor and fingerprint generation → QSAR/machine learning → computational validation → experimental validation, providing a more robust workflow for computational drug discovery.

14. Strengths, Limitations, and Ethical Considerations

Strengths

- The project demonstrates a structured Python-based workflow for molecular data visualization and exploratory analysis.
- Multiple visualization techniques were used to examine molecular-property distributions, descriptor relationships, and chemical space.
- The analysis connects molecular descriptors with concepts relevant to medicinal chemistry, QSAR, and computational drug discovery.
- The use of a controlled synthetic dataset provides a suitable environment for demonstrating data-analysis techniques.

Limitations

- The dataset was synthetically generated and does not represent experimentally validated chemical or biological data.
- The bioactivity labels were not experimentally measured, limiting the biological interpretation of the results.
- Only six molecular descriptors were analysed, providing a limited representation of molecular structure.
- The observed correlations and differences between Active and Inactive compounds cannot be considered experimentally validated structure–activity relationships.
- The analysis does not include experimental validation or external validation using an independent chemical dataset.

Ethical Considerations

Computational and AI-based approaches in drug discovery should be used responsibly. Predictions should not be presented as experimentally confirmed biological outcomes without appropriate validation. Models developed from limited or synthetic datasets may produce misleading conclusions when applied to real-world chemical and biological data.

Therefore, data quality, transparent reporting, appropriate validation, reproducibility, and responsible interpretation are important when applying machine learning and AI to pharmaceutical research. Computational predictions should complement rather than replace experimental investigation.

Overall, recognizing these strengths, limitations, and ethical considerations is essential for the responsible application of data science and AI in computational drug discovery.

15. Conclusion

This project demonstrated the application of data visualization and exploratory data analysis to molecular descriptor data in the context of computational drug discovery. A synthetic dataset of 500 chemical compounds was analysed using Molecular Weight, LogP, HBA, HBD, TPSA, and Rotatable Bonds.

Various visualization techniques were used to examine molecular-property distributions, relationships between descriptors, differences between Active and Inactive compounds, correlations, chemical space, and potential outliers. The analysis showed substantial overlap between the Active and Inactive groups across several molecular properties, highlighting the importance of considering multiple molecular features rather than relying on individual descriptors.

From a medicinal chemistry perspective, the project provided an understanding of how physicochemical properties can be explored and interpreted within a chemical dataset. It also demonstrated how exploratory analysis can serve as an initial step toward QSAR, machine learning, molecular fingerprint analysis, and AI-assisted drug discovery.

However, the synthetic nature of the dataset and the absence of experimentally validated bioactivity data limit the biological conclusions that can be drawn. Therefore, the findings should be considered exploratory rather than experimentally validated.

Overall, the project provided practical experience in Python-based data visualization, molecular descriptor analysis, chemical-space exploration, and computational drug discovery workflows, establishing a foundation for more advanced studies using real-world chemical and biological datasets.

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17. Appendix

The appendix contains the Python code and supporting screenshots used to generate the analyses and visualizations presented in the report. It provides additional technical details while keeping the main report focused on the interpretation of results.

The appendix can be organized as follows:

Appendix A – Dataset Generation and Preparation

This appendix presents the Python code used to generate the synthetic chemical dataset and perform the initial data inspection and preparation. The dataset contains 500 chemical compounds represented using Molecular Weight (MW), LogP, HBA, HBD, TPSA, Rotatable Bonds (RB), and Bioactivity classification.

A.1 Synthetic Dataset Generation

```
import numpy as np

import pandas as pd


# Set random seed for reproducibility
np.random.seed(42)


# Number of synthetic compounds
n = 500


# Generate synthetic chemical dataset
df = pd.DataFrame({
    "Compound_ID": [f"CMP{i+1}" for i in range(n)],
    "Molecular_Weight": np.random.normal(350, 50, n),
    "LogP": np.random.normal(2.5, 1.0, n),
```

```
"HBA": np.random.randint(1, 10, n),  
"HBD": np.random.randint(0, 6, n),  
"TPSA": np.random.normal(70, 20, n),  
"Rotatable_Bonds": np.random.randint(0, 10, n),  
"Bioactivity": np.random.randint(0, 2, n)  
})
```

```
# Convert binary values into bioactivity labels  
df["Bioactivity_Class"] = df["Bioactivity"].map({  
    0: "Inactive",  
    1: "Active"  
})
```

```
# Display the first five records  
print(df.head())
```

A.2 Data Inspection

```
# Display dataset dimensions  
print("Dataset shape:", df.shape)
```

```
# Display column names  
print("\nColumn names:")  
print(df.columns.tolist())
```

```
# Display data types
```

```
print("\nData types:")
```

```
print(df.dtypes)
```

```
# Check for missing values
```

```
print("\nMissing values:")
```

```
print(df.isnull().sum())
```

```
# Check for duplicate rows
```

```
print("\nNumber of duplicate rows:", df.duplicated().sum())
```

```
# Display statistical summary
```

```
print("\nStatistical summary:")
```

```
print(df.describe())
```

A.3 Molecular Descriptor Selection

```
# Select molecular descriptors used in the analysis
```

```
descriptors = [
```

```
    "Molecular_Weight",
```

```
    "LogP",
```

```
    "HBA",
```

```
    "HBD",
```

```
    "TPSA",
```

```
    "Rotatable_Bonds"
```

```
]
```



```
print("Selected molecular descriptors:")
```

A.5

```
print(descriptors)
```

A.4 Data Preparation

```
# Confirm descriptor data types
```

```
print("\nDescriptor data types:")
```

```
print(df[descriptors].dtypes)
```

```
# Check for missing values in analysis variables
```

```
print("\nMissing values:")
```

```
print(df[descriptors + ["Bioactivity_Class"]].isnull().sum())
```

```
# Display bioactivity class distribution
```

```
print("\nBioactivity class distribution:")
```

```
print(df["Bioactivity_Class"].value_counts())
```

A.5 Purpose of the Code

The above code establishes a reproducible synthetic dataset for the subsequent molecular descriptor analysis, exploratory visualization, Active vs Inactive comparison, correlation analysis, chemical-space analysis, and outlier analysis presented in this report.

Since the dataset is synthetically generated, the values are intended for demonstration and exploratory analysis and do not represent experimentally measured molecular or biological data.

Appendix B - Molecular Descriptor Analysis

This appendix presents the Python code used to select, summarize, and analyze the molecular descriptors used throughout the project. The analysis focuses on Molecular Weight (MW), LogP, HBA, HBD, TPSA, and Rotatable Bonds (RB).

B.1 Molecular Descriptor Selection

```
# Select molecular descriptors for analysis
```

```
descriptors = [  
    "Molecular_Weight",  
    "LogP",  
    "HBA",  
    "HBD",  
    "TPSA",  
    "Rotatable_Bonds"  
]
```

```
# Display selected descriptors
```

```
print("Molecular descriptors used in the analysis:")
```

```
for descriptor in descriptors:
```

```
    print("-", descriptor)
```

B.2 Descriptive Statistics

Generate descriptive statistics for molecular descriptors

```
descriptor_summary = df[descriptors].describe()
```

```
print("Descriptive statistics of molecular descriptors:")
```

```
print(descriptor_summary)
```

B.3 Mean and Median Analysis

Calculate mean and median values

```
mean_values = df[descriptors].mean()
```

```
median_values = df[descriptors].median()
```

```
print("Mean values:")
```

```
print(mean_values)
```

```
print("\nMedian values:")
```

```
print(median_values)
```

B.4 Descriptor Range Analysis

Calculate minimum and maximum values

```
minimum_values = df[descriptors].min()
```

```
maximum_values = df[descriptors].max()
```

```
print("Minimum values:")
```

```
print(minimum_values)
```

```
print("\nMaximum values:")
```

```
print(maximum_values)
```

B.5 Summary of Descriptor Properties

```
# Create a compact summary table
```

```
summary_table = pd.DataFrame({  
    "Mean": df[descriptors].mean(),  
    "Median": df[descriptors].median(),  
    "Minimum": df[descriptors].min(),  
    "Maximum": df[descriptors].max(),  
    "Standard Deviation": df[descriptors].std()  
})
```

```
print(summary_table)
```

B.6 Purpose of the Code

The code in this appendix was used to obtain the basic statistical characteristics of the molecular descriptors before performing visualization and comparative analysis. These statistics provide information about the central tendency, variability, and range of the molecular-property data.

The descriptor analysis supports the subsequent sections of the report, including exploratory visualization, Active vs Inactive comparison, correlation analysis, chemical-space interpretation, and outlier analysis.

Appendix C – Exploratory Data Visualization

This appendix presents the Python code used to generate the visualizations in Section 6: Exploratory Data Visualization. The analysis includes distributions of molecular properties, bioactivity class distribution, and relationships between selected molecular descriptors.

C.1 Molecular Weight Distribution

```
import matplotlib.pyplot as plt

import seaborn as sns

plt.figure(figsize=(8, 5))

sns.histplot(
    data=df,
    x="Molecular_Weight",
    bins=30,
    kde=True
)

plt.title("Distribution of Molecular Weight")
plt.xlabel("Molecular Weight (Da)")
plt.ylabel("Number of Compounds")
plt.tight_layout()
plt.show()
```

C.2 LogP Distribution

```
plt.figure(figsize=(8, 5))

sns.histplot(
    data=df,
    x="LogP",
    bins=30,
    kde=True
)

plt.title("Distribution of LogP")
plt.xlabel("LogP")
plt.ylabel("Number of Compounds")
plt.tight_layout()
plt.show()
```

C.3 Bioactivity Class Distribution

```
plt.figure(figsize=(7, 5))

sns.countplot(
    data=df,
    x="Bioactivity_Class"
)
```

```
plt.title("Distribution of Active and Inactive Compounds")  
plt.xlabel("Bioactivity Class")  
plt.ylabel("Number of Compounds")  
plt.tight_layout()  
plt.show()
```

C.4 Molecular Weight vs LogP

```
plt.figure(figsize=(8, 6))  
  
sns.scatterplot(  
    data=df,  
    x="Molecular_Weight",  
    y="LogP",  
    hue="Bioactivity_Class"  
)
```

```
plt.title("Molecular Weight vs LogP")  
plt.xlabel("Molecular Weight (Da)")  
plt.ylabel("LogP")  
plt.tight_layout()  
plt.show()
```

C.5 TPSA vs LogP

```
plt.figure(figsize=(8, 6))
```

```
sns.scatterplot(  
    data=df,  
    x="TPSA",  
    y="LogP",  
    hue="Bioactivity_Class"  
)  
  
plt.title("TPSA vs LogP")  
plt.xlabel("Topological Polar Surface Area (Å²)")  
plt.ylabel("LogP")  
plt.tight_layout()  
plt.show()
```

C.6 Purpose of the Code

The visualization code was used to examine molecular-property distributions, bioactivity class distribution, and relationships between selected molecular descriptors. These visualizations provide an initial understanding of the chemical-property space and support the subsequent comparative, correlation, and medicinal chemistry analyses.

Appendix D - Active vs Inactive Compound Comparison

This appendix presents the Python code used to generate the visualization in section 7.1, which compares the distributions of Molecular Weight (MW), LogP, HBA, HBD, TPSA, and Rotatable Bonds (RB) between Active and Inactive compounds.

D.1 Box Plot Comparison

```
import matplotlib.pyplot as plt

import seaborn as sns

# Molecular descriptors
descriptors = [
    "Molecular_Weight",
    "LogP",
    "HBA",
    "HBD",
    "TPSA",
    "Rotatable_Bonds"
]

# Create box plots for Active vs Inactive compounds
fig, axes = plt.subplots(2, 3, figsize=(15, 9))

for ax, descriptor in zip(axes.flatten(), descriptors):
    sns.boxplot(
        data=df,
        x="Bioactivity_Class",
        y=descriptor,
        ax=ax
    )
```

```
ax.set_title(f"{descriptor} by Bioactivity")
ax.set_xlabel("Bioactivity Class")
ax.set_ylabel(descriptor)

plt.suptitle(
    "Comparison of Molecular Descriptors Between Active and Inactive
    Compounds",
    fontsize=14
)

plt.tight_layout()
plt.show()
```

D.2 Purpose of the Code

The box plots provide a visual comparison of the central tendency, spread, and potential outliers of molecular descriptors between Active and Inactive compounds.

This analysis supports the discussion in Section 7: Comparative Analysis of Active and Inactive Compounds and helps determine whether individual molecular properties show noticeable differences between the two bioactivity classes.

Because the dataset is synthetic, the observed differences are considered exploratory patterns rather than experimentally validated structure–activity relationships.

Appendix E - Correlation and Chemical-Space Analysis

This appendix presents the Python code used to generate Figure: Correlation Heatmap and Figure: Pairwise Descriptor Analysis. These visualizations were used to examine relationships among the selected molecular descriptors and explore the chemical-property space.

E.1 Correlation Heatmap

```
import matplotlib.pyplot as plt

import seaborn as sns

# Select molecular descriptors
descriptors = [
    "Molecular_Weight",
    "LogP",
    "HBA",
    "HBD",
    "TPSA",
    "Rotatable_Bonds"
]

# Calculate correlation matrix
correlation_matrix = df[descriptors].corr()

# Create heatmap
plt.figure(figsize=(9, 7))

sns.heatmap(
```

```
correlation_matrix,  
annot=True,  
cmap="coolwarm",  
fmt=".2f",  
linewidths=0.5  
)  
  
plt.title("Correlation Heatmap of Molecular Descriptors")  
plt.tight_layout()  
plt.show()
```

E.2 Pairwise Descriptor Analysis

Create pairwise plot of molecular descriptors

```
sns.pairplot(  
    df,  
    vars=descriptors,  
    hue="Bioactivity_Class",  
    diag_kind="hist"  
)  
  
plt.suptitle(  
    "Pairwise Relationships Among Molecular Descriptors",  
    y=1.02  
)
```

```
plt.show()
```

E.3 Purpose of the Code

The correlation heatmap was used to identify the strength and direction of relationships among molecular descriptors, while the pairwise plot provided a visual representation of their distributions and relationships.

These analyses support Section 8: Correlation and Chemical-Space Analysis and help in understanding descriptor relationships before applying QSAR, machine learning, or AI-based approaches.

Because the dataset is synthetic, the observed correlations should be considered exploratory and not experimentally validated.

Appendix F - Outlier Analysis

This appendix presents the Python code used to identify and visualize potential outliers in the molecular descriptor dataset. The analysis focuses on Molecular Weight (MW), LogP, HBA, HBD, TPSA, and Rotatable Bonds (RB).

F.1 Box Plot-Based Outlier Visualization

The box plots generated in Figure: Distribution of Molecular Properties by Bioactivity were also used to visually identify observations that fall outside the typical range of each molecular descriptor.

```
import matplotlib.pyplot as plt  
  
import seaborn as sns
```

```
# Molecular descriptors  
descriptors = [  
    "Molecular_Weight",  
    "LogP",
```

```
"HBA",
"HBD",
"TPSA",
"Rotatable_Bonds"
]

# Create box plots for outlier visualization
fig, axes = plt.subplots(2, 3, figsize=(15, 9))

for ax, descriptor in zip(axes.flatten(), descriptors):
    sns.boxplot(
        data=df,
        y=descriptor,
        ax=ax
    )

    ax.set_title(f"Outlier Analysis: {descriptor}")
    ax.set_ylabel(descriptor)

plt.suptitle(
    "Outlier Analysis of Molecular Descriptors",
    fontsize=14
)

plt.tight_layout()
```

```
plt.show()
```

F.2 IQR-Based Outlier Identification

The Interquartile Range (IQR) method was used to identify potential numerical outliers.

```
# Identify potential outliers using the IQR method
```

```
outlier_summary = {}
```

```
for descriptor in descriptors:
```

```
    Q1 = df[descriptor].quantile(0.25)
```

```
    Q3 = df[descriptor].quantile(0.75)
```

```
    IQR = Q3 - Q1
```

```
    lower_bound = Q1 - 1.5 * IQR
```

```
    upper_bound = Q3 + 1.5 * IQR
```

```
    outliers = df[
```

```
        (df[descriptor] < lower_bound) |
```

```
        (df[descriptor] > upper_bound)
```

```
    ]
```

```
    outlier_summary[descriptor] = len(outliers)
```

```
print(f"\n{descriptor}")  
print("Lower bound:", round(lower_bound, 2))  
print("Upper bound:", round(upper_bound, 2))  
print("Number of potential outliers:", len(outliers))
```

F.3 Outlier Summary

Convert outlier results into a summary table

```
outlier_table = pd.DataFrame(  
    list(outlier_summary.items()),  
    columns=["Molecular Descriptor", "Potential Outliers"]  
)  
  
print(outlier_table)
```

F.4 Purpose of the Code

The code was used to identify molecular descriptor values that fall outside the expected range according to the $1.5 \times \text{IQR}$ criterion. The analysis provides additional information about the variability and distribution of the chemical-property data.

Potential outliers were not automatically removed, because an unusual molecular property does not necessarily represent an error. Such observations may correspond to compounds with genuinely different physicochemical characteristics.

The results support Section 9: Outlier Analysis and contribute to the interpretation of the chemical-property space from a medicinal chemistry and computational drug discovery perspective.

Appendix G - Complete Visualization Workflow

This appendix presents the Python code used to combine the major visualization steps performed in the project. It provides a consolidated workflow for generating the molecular-property distributions, bioactivity comparison, correlation analysis, and chemical-space visualizations discussed in the main report.

G.1 Required Libraries

```
import pandas as pd  
  
import matplotlib.pyplot as plt  
  
import seaborn as sns
```

G.2 Molecular Descriptor Selection

```
descriptors = [  
    "Molecular_Weight",  
    "LogP",  
    "HBA",  
    "HBD",  
    "TPSA",  
    "Rotatable_Bonds"  
]
```

G.3 Distribution of Molecular Descriptors

```
fig, axes = plt.subplots(2, 3, figsize=(15, 9))
```

```
for ax, descriptor in zip(axes.flatten(), descriptors):
```

```
    sns.histplot(
        data=df,
        x=descriptor,
        bins=30,
        kde=True,
        ax=ax
    )
```

```
    ax.set_title(f"Distribution of {descriptor}")
```

```
    ax.set_xlabel(descriptor)
```

```
    ax.set_ylabel("Number of Compounds")
```

```
plt.tight_layout()
```

```
plt.show()
```

G.4 Active vs Inactive Comparison

```
fig, axes = plt.subplots(2, 3, figsize=(15, 9))
```

```
for ax, descriptor in zip(axes.flatten(), descriptors):
```

```
    sns.boxplot(
        data=df,
```

```
x="Bioactivity_Class",  
y=descriptor,  
ax=ax  
)  
  
ax.set_title(f"{descriptor} by Bioactivity")  
ax.set_xlabel("Bioactivity Class")  
ax.set_ylabel(descriptor)
```

```
plt.suptitle(  
    "Comparison of Molecular Descriptors Between Active and Inactive  
Compounds",  
    fontsize=14  
)
```

```
plt.tight_layout()  
plt.show()
```

G.5 Correlation Heatmap

```
correlation_matrix = df[descriptors].corr()
```

```
plt.figure(figsize=(9, 7))
```

```
sns.heatmap(  
    correlation_matrix,
```

```
annot=True,  
cmap="coolwarm",  
fmt=".2f",  
linewidths=0.5  
)  
  
plt.title("Correlation Heatmap of Molecular Descriptors")  
plt.tight_layout()  
plt.show()
```

G.6 Pairwise Chemical-Space Analysis

```
sns.pairplot(  
    df,  
    vars=descriptors,  
    hue="Bioactivity_Class",  
    diag_kind="hist"  
)  
  
plt.suptitle(  
    "Pairwise Relationships Among Molecular Descriptors",  
    y=1.02  
)  
  
plt.show()
```

G.7 Purpose of the Code

The consolidated workflow demonstrates how Python-based visualization can be applied to explore molecular-property distributions, bioactivity differences, descriptor correlations, and chemical-property space.

The code supports the major analytical sections of the project and demonstrates the use of Pandas, Matplotlib, and Seaborn for molecular data analysis and visualization.